

PRATHAM RAJ

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EDUCATION

National Institute of Technology Patna

B.Tech in Electrical Engineering

Patna, India

Aug 2023 – May 2027

TECHNICAL SKILLS

Languages: C/C++, Python, Java, JavaScript (ES6+), TypeScript, HTML5, CSS3

Backend & APIs: Node.js, Express.js, RESTful APIs, JWT Authentication, OAuth 2.0, Socket/Network Programming

Frontend: React.js, Zustand, Redux, Tailwind CSS, Vite, Responsive Design, PWA

Databases & Cloud: MongoDB, MySQL, Redis, Cloudinary, Vercel, Render, Mongoose ODM

Tools & Practices: Git/GitHub, Postman, Linux/Unix, CMake, Docker (basic), Agile, Code Review, CI/CD

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design, Multi-threading, Scalability

EXPERIENCE

Web Developer Intern

May 2026 – July 2026

InternPe (Certificate) | [GitHub](#) | [Live](#)

Remote

- Built an Interactive Question Sheet SPA using **React 19** and **Zustand** with a Topic → Subtopic → Question data hierarchy, drag-and-drop reordering (@dnd-kit), and real-time progress tracking.
- Designed accessible, responsive UI components with **Tailwind CSS**; deployed on **Vercel** with `localStorage` persistence and **Firebase** authentication for cross-device sync.

PROJECTS

VectorForge – C++ Vector Database & RAG Engine | *C++17, HNSW, KD-Tree, REST API, Ollama* | [GitHub](#)

- Implemented three interchangeable K-NN search algorithms from scratch (**Brute-Force, KD-Tree, HNSW**) achieving $O(\log N)$ approximate search via a custom multi-layer HNSW graph ($M=16, e_{search}=50$) – the same algorithm powering Pinecone, Weaviate, and Chroma.
- Built a local **RAG pipeline**: text → Ollama (nomic-embed-text 768D embeddings) → HNSW semantic search → Llama 3.2 generation, with 250-word overlapping chunking and transactional rollback on partial embedding failures.
- Exposed a **15-route REST API** via `cpp-http-lib` with mutex-guarded thread-safe stores, input validation, and a client-side **PCA scatter-plot** for interactive nearest-neighbor visualization.

NIT Patna Market – Campus Marketplace | *React.js, Node.js, MongoDB, Cloudinary, JWT* | [GitHub](#) | [Live](#)

- Led full-stack **MERN** development with **JWT authentication**, OTP-verified `@nitp.ac.in` signup (Resend API), and **role-based access control** across guest/user/seller/admin tiers.
- Built dual-mode **image pipeline** (Cloudinary CDN + local-disk fallback) for up to 8 images/listing with cascade cleanup; buyer-seller **HTTP polling chat** with read receipts and cron-scheduled email digest.
- Deployed as a **PWA** on Vercel + Render; accumulated 2 GitHub forks and active student usage across NIT Patna campus.

NetSentry – Deep Packet Inspection Engine | *C++17, libpcap, Multi-threading, TLS/SNI, CMake* | [GitHub](#)

- Built a **multi-threaded DPI engine** in C++ parsing raw PCAP captures, inspecting TCP/UDP payloads at **Layer 4–7**, and extracting **TLS SNI** fields from encrypted HTTPS traffic to classify apps (YouTube, Facebook, Google).
- Implemented a **producer-consumer pipeline** (Reader → Load Balancer threads → Fast Path workers → Output Writer) with consistent hash-based flow affinity, achieving $O(\log N)$ throughput scaling and 100% targeted-flow blocking accuracy on test PCAP datasets.

ACHIEVEMENTS & COMPETITIVE PROGRAMMING

Amazon HackOn 6.0: Ranked **Top 75 coders** globally out of 70,000+ participants.

Competitive Programming: Knight Badge (2000+ rating) on LeetCode | **3-Star** (1650+) on CodeChef | **Pupil** (1200+) on Codeforces | 1,500+ problems solved.

Hackathon Champion: 1st place, Hackslash College Hackathon – rapid prototyping and algorithmic problem-solving under time constraints.

Industry Recognition: Shortlisted for **Digital Specialist Engineer (DSE)** at Infosys via top-percentile technical assessments. Ranked **Top 10** institution-wide on Codolio leaderboard at NIT Patna.

LEADERSHIP & VOLUNTEERING

Campus Ambassador, HackerRank & GfG: Organized DSA workshops and competitive programming initiatives, mentoring peers in algorithmic thinking.

Educational Volunteer, NSS (Sankalp): Instructed 500+ students through community-driven educational outreach.

Technical Contributor, GDSC: Collaborated on open-source software solutions and developer community initiatives.